

Set Theory

6.1. Basics of Set Theory

6.2 **Properties of Sets and Element Argument**

6.3 Algebraic Proofs

6.4 Boolean Algebras



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Acknowledgement:

This lecture is based on (but not limited to) to chapter 5 in “Discrete Mathematics with Applications by Susanna S. Epp (3rd Edition)”.

Set Theory

6.2 Properties of Sets

In this lecture:



☒ **Part 1: Set Relations and Identities**

☐ Part 2: Proving Set Identities (Element Argument)

☐ Part 3: Examples of proving Set Identities

Set Relations

Theorem 6.2.1 Some Subset Relations

1. *Inclusion of Intersection:* For all sets A and B ,
$$(a) A \cap B \subseteq A \quad \text{and} \quad (b) A \cap B \subseteq B.$$
2. *Inclusion in Union:* For all sets A and B ,
$$(a) A \subseteq A \cup B \quad \text{and} \quad (b) B \subseteq A \cup B.$$
3. *Transitive Property of Subsets:* For all sets A , B , and C ,
$$\text{if } A \subseteq B \text{ and } B \subseteq C, \text{ then } A \subseteq C.$$

Procedural Versions of Set Definitions

Let X and Y be subsets of a universal set U and suppose x and y are elements of U .

1. $x \in X \cup Y \iff x \in X \text{ or } x \in Y$
2. $x \in X \cap Y \iff x \in X \text{ and } x \in Y$
3. $x \in X - Y \iff x \in X \text{ and } x \notin Y$
4. $x \in X^c \iff x \notin X$
5. $(x,y) \in X \times Y \iff x \in X \text{ and } y \in Y$

Theorem 6.2.2 Set Identities

Let all sets referred to below be subsets of a universal set U .

1. *Commutative Laws*: For all sets A and B ,

$$(a) A \cup B = B \cup A \quad \text{and} \quad (b) A \cap B = B \cap A.$$

2. *Associative Laws*: For all sets A , B , and C ,

$$(a) (A \cup B) \cup C = A \cup (B \cup C) \quad \text{and}$$

$$(b) (A \cap B) \cap C = A \cap (B \cap C).$$

3. *Distributive Laws*: For all sets, A , B , and C ,

$$(a) A \cup (B \cap C) = (A \cup B) \cap (A \cup C) \quad \text{and}$$

$$(b) A \cap (B \cup C) = (A \cap B) \cup (A \cap C).$$

4. *Identity Laws*: For all sets A ,

$$(a) A \cup \emptyset = A \quad \text{and} \quad (b) A \cap U = A.$$

5. *Complement Laws*:

$$(a) A \cup A^c = U \quad \text{and} \quad (b) A \cap A^c = \emptyset.$$

6. *Double Complement Law*: For all sets A ,

$$(A^c)^c = A.$$

Theorem 6.2.2 Set Identities

Let all sets referred to below be subsets of a universal set U .

6. *Double Complement Law*: For all sets A ,

$$(A^c)^c = A.$$

7. *Idempotent Laws*: For all sets A ,

$$(a) A \cup A = A \quad \text{and} \quad (b) A \cap A = A.$$

8. *Universal Bound Laws*: For all sets A ,

$$(a) A \cup U = U \quad \text{and} \quad (b) A \cap \emptyset = \emptyset.$$

9. *De Morgan's Laws*: For all sets A and B ,

$$(a) (A \cup B)^c = A^c \cap B^c \quad \text{and} \quad (b) (A \cap B)^c = A^c \cup B^c.$$

10. *Absorption Laws*: For all sets A and B ,

$$(a) A \cup (A \cap B) = A \quad \text{and} \quad (b) A \cap (A \cup B) = A.$$

11. *Complements of U and $\emptyset$* :

$$(a) U^c = \emptyset \quad \text{and} \quad (b) \emptyset^c = U.$$

12. *Set Difference Law*: For all sets A and B ,

$$A - B = A \cap B^c.$$

→ We will prove some of these theories in the lecture, please prove others at home

Set Theory

6.2 Properties of Sets

In this lecture:

☐ Part 1: Set Relations and Identities



☐ Part 2: **Proving Set Identities (Element Argument)**

☐ Part 3: Examples of proving Set Identities

Proving Set Identities

Proving That Sets Are Equal

e.g., prove: $HumanMale = Man$

Basic Method for Proving That Sets Are Equal

Let sets X and Y be given. To prove that $X=Y$:

1. Prove that $X \subseteq Y$.
2. Prove that $Y \subseteq X$.

But: How to prove that $X \subseteq Y$?

The Element Argument Method

For Proving a set is a subset of another

e.g., prove: $HumanMale \subseteq Man$

i.e., Prove that every element in $HumanMale$ is an element in Man

The Element Argument Method:

Let sets X and Y be Given, To Prove that $X \subseteq Y$:

Step 1. Suppose that x is a particular but arbitrarily chosen element of X .

Step 2. Show that x is an element of Y .

The Element Argument Method

In details

Example: Prove that: $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

That is:

Prove: $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$

That is, show $\forall x$, if $x \in A \cup (B \cap C)$ then $x \in (A \cup B) \cap (A \cup C)$

Suppose $x \in A \cup (B \cap C)$. [Show $x \in (A \cup B) \cap (A \cup C)$.]

...

Thus $x \in (A \cup B) \cap (A \cup C)$.

Hence $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$.

Prove: $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$

That is, show $\forall x$, if $x \in (A \cup B) \cap (A \cup C)$ then $x \in A \cup (B \cap C)$.

Suppose $x \in (A \cup B) \cap (A \cup C)$. [Show $x \in A \cup (B \cap C)$.]

...

Thus $x \in A \cup (B \cap C)$.

Hence $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$.

Thus $(A \cup B) \cap (A \cup C) = A \cup (B \cap C)$.

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6.2 Properties of Sets

In this lecture:

- ☐ Part 1: Set Relations and Identities
- ☐ Part 2: Proving Set Identities (Element Argument)
-  ☐ Part 3: **Examples of proving Set Identities**

Proving: A Distributive Law for Sets

Theorem 6.2.2(3)(a) A Distributive Law for Sets

For all sets A, B, and C,

$$A \cup (B \cap C) = (A \cup B) \cap (A \cup C).$$

$$A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C):$$

Suppose $x \in A \cup (B \cap C)$.

$x \in A$ or $x \in B \cap C$. (by def. of union)

Case 1 ($x \in A$): then

$x \in A \cup B$ (by def. of union) and

$x \in A \cup C$ (by def. of union)

$\therefore x \in (A \cup B) \cap (A \cup C)$ (def. of intersection)

Case 2 ($x \in B \cap C$): then

$x \in B$ and $x \in C$ (def. of intersection)

As $x \in B$, $x \in A \cup B$ (by def. of union)

As $x \in C$, $x \in A \cup C$, (by def. of union)

$\therefore x \in (A \cup B) \cap (A \cup C)$ (def. of intersection)

In both cases, $x \in (A \cup B) \cap (A \cup C)$.

Thus: $A \cup (B \cap C) \subseteq (A \cup B) \cap (A \cup C)$

by definition of subset

$$(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C):$$

Suppose $x \in (A \cup B) \cap (A \cup C)$.

$x \in A \cup B$ and $x \in A \cup C$. (def. of intersection)

Case 1 ($x \in A$): then

$x \in A \cup (B \cap C)$ (by def. of union)

Case 2 ($x \notin A$): then

$x \in B$ and $x \in C$, (def. of intersection)

Then, $x \in B \cap C$ (def. of intersection)

$\therefore x \in A \cup (B \cap C)$

In both cases $x \in A \cup (B \cap C)$.

Thus: $(A \cup B) \cap (A \cup C) \subseteq A \cup (B \cap C)$

by definition of subset,

Conclusion: Since both subset relations have been proved, it follows by definition of set equality that $A \cup (B \cap C) = (A \cup B) \cap (A \cup C)$.

Proving: A De Morgan's Law for Sets

Theorem 6.2.2(9)(a) A De Morgan's Law for Sets

For all sets A and B , $(A \cup B)^c = A^c \cap B^c$

Same As: proving whether: the people who are not students or employees is the same as the people who are either students nor employees.

$$(A \cup B)^c \subseteq A^c \cap B^c$$

Suppose $x \in (A \cup B)^c$. [We must show that $x \in A^c \cap B^c$.] By definition of complement,

$$x \notin A \cup B.$$

But to say that $x \notin A \cup B$ means that

it is false that $(x \text{ is in } A \text{ or } x \text{ is in } B)$.

By De Morgan's laws of logic, this implies that

x is not in A and x is not in B ,

which can be written $x \notin A$ and $x \notin B$.

Hence $x \in A^c$ and $x \in B^c$ by definition of complement. It follows, by definition of intersection, that $x \in A^c \cap B^c$ [as was to be shown]. So $(A \cup B)^c \subseteq A^c \cap B^c$ by definition of subset.

$$A^c \cap B^c \subseteq (A \cup B)^c$$

Suppose $x \in A^c \cap B^c$. [We must show that $x \in (A \cup B)^c$.] By definition of intersection, $x \in A^c$ and $x \in B^c$, and by definition of complement,

$$x \notin A \quad \text{and} \quad x \notin B.$$

In other words, x is not in A and x is not in B .

By De Morgan's laws of logic this implies that

it is false that $(x \text{ is in } A \text{ or } x \text{ is in } B)$,

which can be written $x \notin A \cup B$

by definition of union. Hence, by definition of complement, $x \in (A \cup B)^c$ [as was to be shown]. It follows that $A^c \cap B^c \subseteq (A \cup B)^c$ by definition of subset.

Theorem 6.2.3 Intersection and Union with a Subset

For any sets A and B , if $A \subseteq B$, then

$$(a) A \cap B = A \quad \text{and} \quad (b) A \cup B = B.$$

Prove at home

Proof: If every person is a student, then the set of persons and students are students

Part (a): Suppose A and B are sets with $A \subseteq B$. To show part (a) we must show both that $A \cap B \subseteq A$ and that $A \subseteq A \cap B$. We already know that $A \cap B \subseteq A$ by the inclusion of intersection property. To show that $A \subseteq A \cap B$, let $x \in A$. [We must show that $x \in A \cap B$.] Since $A \subseteq B$, then $x \in B$ also. Hence

$$x \in A \quad \text{and} \quad x \in B,$$

and thus

$$x \in A \cap B$$

by definition of intersection [as was to be shown].

Theorem 6.2.4 A Set with No Elements Is a Subset of Every Set

If E is a set with no elements and A is any set, then $E \subseteq A$.

المجموعة التي لا تحتوي على أي عنصر
(E المجموعة الخالية)،
في الكون A هي مجموعة جزئية من أي مجموعة أخرى

Proof by Contradiction:

Suppose not. [We take the negation of the theorem and suppose it to be true.]

That is, Suppose: E with no elements, and $E \not\subseteq A$.

assuming ($E \not\subseteq A$) means there $x \in E$ and this $x \notin A$ [by definition of subset].

But there can be no such element since E has no elements. **This is a contradiction.**

Hence the supposition that there are sets E and A , where E has no elements and $E \not\subseteq A$, is false, and so the theorem is true.

Proving: Uniqueness of the Empty Set

Corollary 6.2.5 Uniqueness of the Empty Set

There is only one set with no elements.

Proof:

Suppose E_1 and E_2 are both sets with no elements.

By Theorem 6.2.4, $E_1 \subseteq E_2$ since E_1 has no elements.

Also $E_2 \subseteq E_1$ since E_2 has no elements.

Thus $E_1 = E_2$ by definition of set equality.

Proving: a Conditional Statement

Example: If every student is smart and every smart is not-foolish, then there are no foolish students

Proposition 6.2.6

For all sets A , B , and C , if $A \subseteq B$ and $B \subseteq C^c$, then $A \cap C = \emptyset$.

Proof:

Suppose not, Suppose there is an element x in $A \cap C$.

Then $x \in A$ and $x \in C$ (By definition of intersection).

As $A \subseteq B$ then $x \in B$ (by definition of subset).

Also, as $B \subseteq C^c$, then $x \in C^c$ (by definition of subset).

So, $x \notin C$ (by definition of complement)

Thus, $x \in C$ and $x \notin C$, which is a contradiction.

So the supposition that there is an element x in $A \cap C$ is false,
and thus $A \cap C = \emptyset$ [as was to be shown].